

Group - 4

Topic Proposals for Milestone 3

Proposal 1:

Title : HIKU: Pull-Based Scheduling for Serverless Computing

Abstract: Serverless computing promises convenient abstractions for developing and deploying functions that execute in response to events. In such Function-as-a-Service (FaaS) platforms, scheduling is an integral task, but current scheduling algorithms often struggle with maintaining balanced loads, minimizing cold starts, and adapting to commonly occurring bursty workloads. In this work, we propose pull-based scheduling as a novel scheduling algorithm for serverless computing. Our key idea is to decouple worker selection from task assignment, with idle workers requesting new tasks proactively. Experimental evaluation on an open-source FaaS platform shows that pull-based scheduling, compared to other existing scheduling algorithms, significantly improves the performance and load balancing of serverless workloads, especially under high concurrency. The proposed algorithm improves response latencies by 14.9% compared to hash-based scheduling, reduces the frequency of cold starts from 43% to 30%, increases throughput by 8.3%, and achieves a more even load distribution by 12.9% measured by the requests assigned per worker.

Year: 21 Feb 2025

Journal: arxiv

Authors: Saman Akbari , Manfred Hauswirth

URL : <https://arxiv.org/abs/2502.15534>

Proposal 2 :

Title : Confidential Serverless Computing (Hacher)

Abstract : Although serverless computing offers compelling cost and deployment simplicity advantages, a significant challenge remains in securely managing sensitive data as it flows through the network of ephemeral function executions in serverless computing environments within untrusted clouds. While Confidential Virtual Machines (CVMs) offer a promising secure execution environment, their integration with serverless architectures currently faces fundamental limitations in key areas: security, performance, and resource efficiency. We present Hacher, a confidential computing system for secure serverless deployments to overcome these limitations.

By employing nested confidential execution and a decoupled guest OS within CVMs, Hacher runs each function in a minimal "trustlet", significantly improving security through a reduced Trusted Computing Base (TCB). Furthermore, by leveraging a data-centric I/O architecture built upon a lightweight LibOS, Hacher optimizes network communication to address performance and resource efficiency challenges. Our evaluation shows that compared to CVM-based deployments, Hacher has $4.3\times$ smaller TCB, improves end-to-end latency (15–93%), achieves higher function density (up to $907\times$), and reduces inter-function communication (up to $27\times$) and function chaining latency ($16.7\text{--}30.2\times$); thus, Hacher offers a practical system for confidential serverless computing.

Year : 1 May 2025

Journal : arxiv

Author : Patrick Sabanic , Masanori Misono , Teofil Bodea , Julian Pritzi , Michael Hackl
Dimitrios Stavrakakis , Pramod Bhatotia

URL : <https://arxiv.org/abs/2504.21518>

Proposal 3:

HAS-GPU

Title : HAS-GPU: Efficient Hybrid Auto-scaling with Fine-grained GPU Allocation for SLO-aware Serverless Inferences

Abstract : Serverless Computing (FaaS) has become a popular paradigm for deep learning inference due to the ease of deployment and pay-per-use benefits. However, current serverless inference platforms encounter the coarse-grained and static GPU resource allocation problems during scaling, which leads to high costs and Service Level Objective (SLO) violations in fluctuating workloads. Meanwhile, current platforms only support horizontal scaling for GPU inferences, thus the cold start problem further exacerbates the problems. In this paper, we propose HAS-GPU, an efficient Hybrid Auto-scaling Serverless architecture with fine-grained GPU allocation for deep learning inferences. HAS-GPU proposes an agile scheduler capable of allocating GPU Streaming Multiprocessor (SM) partitions and time quotas with arbitrary granularity and enables significant vertical quota scalability at runtime. To resolve performance uncertainty introduced by massive fine-grained resource configuration spaces, we propose the Resource-aware Performance Predictor (RaPP). Furthermore, we present an adaptive hybrid auto-scaling algorithm with both horizontal and vertical scaling to ensure inference SLOs and minimize GPU costs. The experiments demonstrated that compared to the mainstream serverless inference platform, HAS-GPU reduces function costs by an average of $10.8\times$ with better SLO guarantees. Compared to state-of-the-art spatio-temporal GPU sharing serverless framework, HAS-GPU reduces function SLO violation by $4.8\times$ and cost by $1.72\times$ on average.

Year : 2 May 2025

Journal: Euro-par

Author : Jianfeng Gu, Puxuan Wang, Isaac David Núñez Araya, Kai Huang and Michael Gerndt

URL : : <https://www.ce.cit.tum.de/caps/aktuelles/news-single-view/article/has-gpu-efficient-hybrid-auto-scaling-with-fine-grained-gpu-allocation-for-slo-aware-serverless-inferences-got-accepted-at-euro-par-2025/>